9.8 Nobody Gets Their Own Shirt

WORKSHEET

The question

STEP II 2016, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s2-q12

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Pause and try Try $n = 3$: list all six orders of cards $1, 2, 3$ and count how many have no card in its own place. Then try $n = 4$, if you are patient (there are 24 orders).

STAGE 1 Inclusion–exclusion for three events

Covers: three events, and four events

Ask yourself Treat $A \cup B$ as a single event and use the two-event formula with C. You will need $(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$.

STAGE 2 The probabilities

Covers: Parts (i) to (iii)

Ask yourself If card i is in position i, in how many ways can the other cards be arranged? And if cards i and j are both in their own positions?

STAGE 3 Adding it up

Covers: at least one card in its own position

Pause and try In the formula for $\mathrm{P}(E_1 \cup \cdots \cup E_n)$, how many terms involve intersections of k events? Multiply by the probability of each, and simplify. Almost everything cancels.

STAGE 4 Exactly one

Covers: exactly one card in its own position

Ask yourself Choose which card is in its own position. What must the other $n-1$ cards then do? Use your previous answer with $n-1$ cards.

What's really going on

Where e comes from

The series $\sum(-1)^k/k!$ is the exponential series $e^x = \sum x^k/k!$ at $x=-1$. So as $n \to \infty$,

$$d_n \to e^{-1} \approx 0.3679,$$

and the probability of exactly one match tends to the same limit. The convergence is extraordinarily fast (Figure 9.8.1): the error in stopping at $1/n!$ is less than $1/(n+1)!$, so already for $n=7$ the answer agrees with $1/e$ to four decimal places. In fact the number of derangements of n objects, $n!\,d_n$, is exactly the nearest integer to $n!/e$.

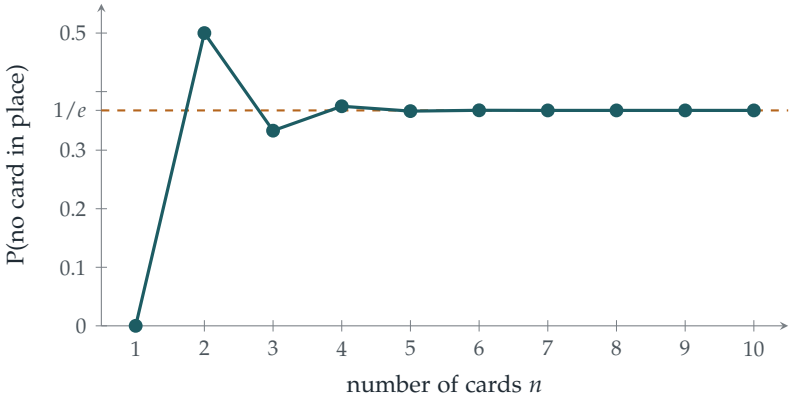


Figure 9.8.1 The probability that a random shuffle of n cards leaves no card in its own place. It alternates above and below $1/e$, and from $n=6$ onwards it is indistinguishable from $1/e$ on this scale.

The number of matches is Poisson

More is true. Let N be the number of cards in their own place. Then $\mathrm{P}(N = k) \to \frac{e^{-1}}{k!}$ for each k: the number of matches has, in the limit, the *Poisson distribution* with mean 1. You can see why the mean is 1 without any calculation: N is the sum of the indicators of $E_1, \ldots, E_n$, each with expectation $\frac{1}{n}$, and expectation adds even though the events are not independent. STEP III 2004 Q12 (in the exercises) shows that the variance is 1 too. The Poisson distribution appears whenever there are many unlikely events that are nearly independent; derangements are a classic example, and the subject of Section 12.11 in Chapter 12.

Montmort's game of *treize*

Pierre Rémond de Montmort, a French nobleman, studied games of chance in his *Essay d'analyse sur les jeux de hazard* (1708). In the game of *treize* (thirteen), a player deals thirteen cards, calling out "one, two, three, ..." and wins if the card dealt ever matches the number called. Montmort found the probability of winning, and in correspondence with Nicolaus Bernoulli the general formula was established by 1713. Euler rediscovered it in 1753. The inclusion–exclusion principle was later formalised by Abraham de Moivre and, in its general form, by James Joseph Sylvester in the nineteenth century.

Try this next

1. **Shirts.** Try STEP III 2004, Question 12 (step.maths.org/questions/04-s3-q12). It counts the players wearing their own shirts with indicator variables, finds the mean and variance, and compares the result with a Poisson distribution.

2. **A recurrence.** Let D_n be the number of derangements of n objects. By considering where card 1 goes, and whether the card in that position goes to position 1, show that $D_n = (n-1)(D_{n-1} + D_{n-2})$. Check it against $D_3 = 2$, $D_4 = 9$.

3. **Exactly k.** Show that the probability that exactly k cards are in their own places is $\frac{1}{k!} d_{n-k}$, and that it tends to $\frac{e^{-1}}{k!}$.

4. **Coprime integers.** Use inclusion–exclusion to count the integers from 1 to 30 divisible by none of 2, 3 and 5. How does your answer relate to $30\left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{3}\right)\left(1 - \frac{1}{5}\right)$?

Further reading

- **Geoffrey Grimmett & Dominic Welsh, *Probability: An Introduction*** (Oxford University Press). A first-year text that covers inclusion–exclusion and the matching problem.
- **William Feller, *An Introduction to Probability Theory and Its Applications*, Vol. 1** (Wiley), Chapter IV. A classic, with many variations on the matching problem.

- **Miklós Bóna, *A Walk Through Combinatorics*** (World Scientific), on inclusion–exclusion.

The full guided solution is in Section 9.8:
https://beyondstep.keithmcnulty.org/ch09/derangements.html